



WOLLO University

Department of software engineering

Exit multiple choice question for

Software project management

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Q1. Software engineering encompasses a ___, a collection of ___ and an array of ___ that allow professionals to build high-quality computer software.

- A. Principles, tools, engineering
- B. Process, methods, tools
- C. Quality, testing, featured
- D. None of the mentioned above

Answer: B) Process, methods, tools

Explanation:

Software engineering encompasses a process, a collection of methods and an array of tools that allow professionals to build high-quality computer software.

Q2. The principles of software engineering are applied during software engineering process by ____ .

- A. Software engineers
- B. Mechanical engineers
- C. Electrical engineers
- D. All of the mentioned above

Answer: A) Software engineers

Explanation:

The principles of software engineering are applied during software engineering process by Software engineers.

Q3. Software engineering enables us to build complex systems in a specified ___ period with high ____.

- A. Analysis, design
- B. Time, quality
- C. Testing, debugging
- D. None of the mentioned above

Answer: B) Time, quality

Explanation:

Software engineering enables us to build complex systems in a specified time period with high quality.

Q4. Software engineering exhibits ___ software.

- A. High quality
- B. Low quality
- C. Medium
- D. None of the mentioned above

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Answer: D) None of the mentioned above

Explanation:

Software engineering exhibits high quality software. It is a layered technology. Any engineering method must be based on a commitment to quality from the organization.

Q5. ____ is a product which designed by software professionals and then support over a long term.

- A. Software
- B. Program
- C. Instructions
- D. None of the mentioned above

Answer: A) Software

Explanation:

Software is a product which designed by software professionals and then support over a long term.

Q6. A Software encompasses ____.

- A. Computer programs
- B. Data structures
- C. Documentation
- D. All of the mentioned above

Answer: D) All of the mentioned above

Explanation:

Software encompasses computer programs, data structures and documentation.

Q7. Software is developed or engineered; it is not ____.

- A. Manufactured
- B. Engineered
- C. Processed
- D. None of the mentioned above

Answer: B) Engineered

Explanation:

Software is developed or engineered; it is not manufactured. Software development is the process of creating and maintaining applications, frameworks, and other software components by conceiving, specifying, designing, programming, documenting, testing, and bug fixing.

Q8. Software is not affected by the same ____ that cause hardware to fail.

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- A. Environmental factors
- B. Development factors
- C. Mechanical process
- D. None of the mentioned above

Answer: A) Environmental factors

Explanation:

Software is not affected by the same environmental factors that cause hardware to fail.

Q9. The term "process" refers to a ___ that must be built in order for software engineering technologies to be delivered effectively.

- A. Framework
- B. Design
- C. Process
- D. None of the mentioned above

Answer: A) Framework

Explanation:

The term "process" refers to a framework that must be built in order for software engineering technologies to be delivered effectively.

Q10. Software engineering ___ provides a way to building software.

- A. Design
- B. Methods
- C. Solutions
- D. Process

Answer: B) Methods

Explanation:

Software engineering design provides a way to building software. Modeling activities and other descriptive techniques are part of software engineering methods, which are based on a set of basic principles that govern each area of technology.

Q11. Software engineering ___ provide automated or semi-automated support for the process and the methods.

- A. Tools
- B. Process
- C. Design
- D. Method

Answer: A) Tools

Explanation:

Software engineering tools provide automated or semi-automated support for the process and the methods. Automated or semi-automated support is provided by software engineering tools when tools are linked together so that data generated by one can be used by another.

Q12. In a software process, an activity refers to achieve,

- A. A broad objective
- B. Communication
- C. Efforts complexity
- D. All of the mentioned above

Answer: D) All of the mentioned above

Explanation:

An activity aims to accomplish a broad goal and can be used regardless of the application domain, project size, effort complexity, or degree of rigor with which software engineering is to be used.

Q13. In software engineering, an architectural design model encompasses a set of tasks.

- A. True
- B. False

Answer: A) True

Explanation:

In software engineering, an architectural design model encompasses a set of tasks. A task focuses on a modest but well-defined goal (for example, running a unit test) that produces a tangible result.

Q14. A process framework establishes a base to complete ____.

- A. Software engineering process
- B. Software testing
- C. Software manufacturing
- D. None of the mentioned above

Answer: A) Software engineering process

Explanation:

A software engineering process, a process framework lays the groundwork for a comprehensive software engineering process by identifying a small number of framework activities that are applicable to all software projects, regardless of their size or complexity.

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Q15. A process of developing a software product using software engineering principles and methods is referred to as ____.

- A. Software design
- B. Software testing
- C. Software evolution
- D. All of the mentioned above

Answer: C) Software evolution

Explanation:

A process of developing a software product using software engineering principles and methods is referred to as software evolution.

Q16. Software evolution process starts from the ____ process.

- A. Design
- B. Analysis
- C. Requirement gathering
- D. Testing

Answer: C) Requirement gathering

Explanation:

Software evolution process starts from the requirement gathering process. After completing this process, a developer creates a prototype of the anticipated software and shows it to consumers to receive input early in the software development process.

Q17. The necessity of software engineering appears because of huge programming.

- A. True
- B. False

Answer: A) True

Explanation:

The necessity of software engineering appears because of huge programming, adaptability, cost, dynamic nature of development and quality management.

Q18. Amongst which of the followings is / are the attributes of good software?

- A. Deliver the required functionality
- B. Performance to the user
- C. Maintainable, dependable and usable
- D. All of the mentioned above

Answer: D) All of the mentioned above

Explanation:

Good software should provide the user with the required functionality and performance while also being maintainable, dependable, and easy to use.

Q19. Amongst which of the followings is / are the Lehman laws for software evolution.

- A. S-type (static-type) - a software, which works according to defined specifications
- B. P-type (practical-type) - software with collection of procedures
- C. E-type (embedded-type) - software which works with real-world environment
- D. All of the mentioned above

Answer: D) All of the mentioned above

Explanation:

The Lehman laws for software evolution are S-type (static-type) - a software, which works according to defined specifications, P-type (practical-type) - software with collection of procedures and E-type (embedded-type) - software which works with real-world environment.

Q20. Software reliability may be considered as an attribute of software quality.

- A. True
- B. False

Answer: A) True

Explanation:

Software reliability is an attribute of software quality. Over an arbitrary time period, the extent to which a program can be expected to perform its desired function.

Q1. Which of the following models is best suited when the requirements of the software are not decided and also the user is not sure about how he wants the user interface to look like?

- a. Classical waterfall model
- b. Prototyping model
- c. Evolutionary model
- d. None of the above

Answer: b. Prototyping model

Explanation:

It is recommended to use prototyping model in the following situations:

- i. Customer requirements are not clear and complete.
- ii. The Design is not clear, i.e. how the software must look like.

Q2. Which of the following software lifecycle model is able to manage all the risks that may appear while software development?

- a. Spiral Waterfall model
- b. Iterative waterfall model
- c. Spiral model
- d. All of the above

Answer: c. Spiral model

Explanation:

The spiral model is a model that has the capability of handling the maximum amount of risks that may occur while developing any software as this model holds the components of almost every other software lifecycle model.

Q3. Which of the following software lifecycle model deals with the incremental development of the software through different versions of the software?

- a. Spiral model
- b. Classical waterfall model
- c. Iterative waterfall model
- d. None of the above

Answer: d. None of the above

Explanation:

The evolutionary model is the one that deals with the incremental development of the software through releasing its different versions.

Q4. Which of the following statements is true?

- i. **In the classical waterfall model, we do not have the scope of risk management as we can only proceed to different phases in the downward manner.**
 - ii. **The iterative model is an enhanced version of the classical waterfall model and it provides us the scope of managing risks in the software.**
- a. Only i is true
 - b. Only ii is true
 - c. Both i and ii are true
 - d. None of them is true

Answer: c. Both i and ii is true

Explanation:

Both the given statements are true.

Q5. In which of the following models is the user feedback considered the most valuable?

- a. Classical waterfall model

- b. Prototyping model
- c. Evolutionary model
- d. None of the above

Answer: b. Prototyping model

Explanation:

The prototyping model follows the build and update cycle until the user is completely satisfied with the prototype of the model. And this happens through the feedback and review that the user gives for the prototype of the software. Thus, in this model, the user feedback is considered of topmost priority.

Q1. Which of the following items are designed in the design phase of the software?

- a. Different modules of the software
- b. Interface between different modules
- c. Data flow and control flow
- d. All of the above

Answer: d. All of the above

Explanation:

In the software design, each component of the software is designed, like the different modules, user interfaces, interfaces between different modules, data flow diagrams, control flow diagrams, etc.

Q2. Which of the following is not a characteristic of a good software design?

- a. Correctness
- b. Understandability
- c. Complex design
- d. Efficiency

Answer: c. Complex design

Explanation:

Good software design must contain the following properties: Correctness, understandability, efficiency, maintainability, etc. However, the complexity of the software design does not ensure it to be a good software design.

Q3. Which of the following objectives are not the one that the software designing phase claim to offer?

- i. **Identify software design activities**
 - ii. **Identify important items developed during the software design phase**
 - iii. **To improve the designing skills of the developers**
- a. All i, ii and iii are correct
 - b. Only i and ii are correct
 - c. Only i and iii are correct

- d. None of the given options is correct

Answer: b. Only i and ii are correct

Explanation:

The iii option is not the objective that the software designing phase aims of.

Q4. Which of the following statements is true?

- i. **The software design phase comes after the feasibility and resources analysis phase.**
 - ii. **The quality of the software depends upon the design phase a lot.**
- a. Only i is true
 - b. Only ii is true
 - c. Both i and ii are true
 - d. None of them is true

Answer: c. Both i and ii are true

Explanation:

Both the given statements are true.

Q5. In which of the following design phases, do the software designers free to make any alterations, corrections and modifications?

- a. Preliminary design phase
- b. Detailed design phase
- c. Both a and b
- d. None of the above

Answer: b. Preliminary design phase

Explanation:

The Preliminary design phase is the one in which the designers propose and study different software designs. They usually show a rough representation of how their software should be like and what things must it contain. Hence, in this phase, the designers are free to do whatever changes they seem useful to make. But, the detailed design phase is the final phase, and not many changes can be made in this phase.

Q1. Understanding requirements in software engineering is,

- A. To understand of what the business impact of the software
- B. What the customer wants
- C. How end users will interact with the software
- D. All of the mentioned above

Answer: D) All of the mentioned above

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Explanation:

Understanding requirements in software engineering is to understand of what the business impact of the software, what the customer wants and how end users will interact with the software.

Q2. In Software Engineering works, amongst which of the following is / are responsible for software requirements,

- A. Software engineers or system engineers
- B. Analysts
- C. Project stakeholders
- D. All of the mentioned above

Answer: D) All of the mentioned above

Explanation:

In Software Engineering works, software engineers or system engineers, Analysts and project stakeholders are responsible for software requirements.

Q3. Why requirement is important in Software Engineering?

- A. To design and build an elegant computer program
- B. To understand what the customer wants
- C. To identify the resource availability for proposed system
- D. All of the mentioned above

Answer: D) All of the mentioned above

Explanation:

Requirement is important in software engineering to design and build an elegant computer program, to understand what the customer wants and to identify the resource availability for proposed system.

Q4. Amongst which of the following is / are the necessary steps for requirements engineering,

- A. To define the scope and nature of the problem
- B. To define what is required
- C. To specify the problem and reviewed or validated to understand the problem
- D. All of the mentioned above

Answer: D) All of the mentioned above

Explanation:

The necessary steps for requirements engineering is to define the scope and nature of the problem, to define what is required and to specify the problem and reviewed or validated to understand the problem.

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Q5. The goal of requirements engineering is to provide a written understanding of the problem to all parties involved.

- A. True
- B. False

Answer: A) True

Explanation:

The goal of requirements engineering is to provide a written understanding of the problem to all parties involved. This can be accomplished through the creation of a variety of work products, including usage scenarios, functions and features lists, requirements models, and specifications.

Q1. Agile software engineering refers,

- A. A philosophy and a set of development guidelines
- B. An active and continuous communication between developers and customers
- C. Encourages customer satisfaction
- D. All of the mentioned above

Answer: D) All of the mentioned above

Explanation:

Agile software engineering refers a philosophy and a set of development guidelines; it is an active and continuous communication between developers and customers and Encourages customer satisfaction.

Q2. In Agile software engineering, agile team is,

- A. Self-organizing and in control of its own destiny
- B. Fosters communication
- C. Collaboration among all who serve on it
- D. All of the mentioned above

Answer: D) All of the mentioned above

Explanation:

In Agile software engineering, agile team is self-organizing and in control of its own destiny; fosters communication and collaboration among all who serve on it.

Q3. Agile software engineering demonstrated to deliver successful systems quickly.

- A. True
- B. False

Answer: A) True

Explanation:

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Agile software engineering demonstrated to deliver successful systems quickly.

Q4. According to Jacobson's view, amongst which of the following is most suitable in terms of the pervasiveness of ___ is the primary driver for agility.

- A. Change
- B. Objective
- C. Team
- D. None of the mentioned above

Answer: A) Change

Explanation:

According to Jacobson's view, amongst which of the following is most suitable in terms of the pervasiveness of change is the primary driver for agility.

Q5. Agility can be applied to any software process using design a process in such a way that,

- A. Allows the project team to adapt tasks
- B. Conduct planning to get the fluidity of an agile development approach
- C. An incremental delivery strategy that gets working software to the customer
- D. All of the mentioned above

Answer: D) All of the mentioned above

Explanation:

Agility can be applied to any software process using design a process that Allows the project team to adapt tasks; conduct planning to get the fluidity of an agile development approach and an incremental delivery strategy that gets working software to the customer.

Q6. An agile software process is characterized in a way that reduces a number of important assumptions about the majority of software project.

- A. True
- B. False

Answer: A) True

Explanation:

An agile software process is characterized in a way that reduces a number of important assumptions about the majority of software project.

Q7. Amongst which of the following is/are the Agility Principles,

- A. To satisfy the customer through early and continuous delivery
- B. Deliver working software frequently
- C. Build projects around motivated individuals
- D. All of the mentioned above

Answer: D) All of the mentioned above

Explanation:

The Agility principles are to satisfy the customer through early and continuous delivery; deliver working software frequently and build projects around motivated individuals.

Q8. Extreme Programming (XP) is an approach of,

- A. Agile software development
- B. Iterative process of software development
- C. Prototype of software development
- D. None of the mentioned above

Answer: A) Agile software development

Explanation:

Extreme Programming (XP) is an approach of agile software development.

Q9. There are ___ phases in Scrum,

- A. Five
- B. Four
- C. Three
- D. None of the mentioned above

Answer: C) Three

Explanation:

There are three phases in Scrum. The first phase consists of an outline planning phase, which is followed by a succession of sprint cycles, and finally, the project closure phase.

Q10. Dynamic systems development method of agile software development which provides a framework for,

- A. Building and maintaining systems
- B. Meets time constraints
- C. Uses of incremental prototyping in a controlled project environment
- D. All of the mentioned above

Answer: D) All of the mentioned above

Explanation:

Dynamic systems development method of agile software development which provides a framework for building and maintaining systems, meets time constraints and uses of incremental prototyping in a controlled project environment.

Q11. Amongst which of the following is/are the type of agile methodologies,

- A. Scrum
- B. FDD
- C. DSDM
- D. All of the mentioned above

Answer: D) All of the mentioned above

Explanation:

Scrum, FDD and DSDM are the type of agile methodologies.

Q12. The core framework activities of Extreme Programming (XP) process are,

- A. Planning, Coding, Testing, Design
- B. Coding, Testing, Planning, Design
- C. Planning, Design, Coding, Testing
- D. Testing, Design, Coding, Planning

Answer: C) Planning, Design, Coding, Testing

Explanation:

The core framework activities of Extreme Programming (XP) process are Planning, Coding, Testing and Design.

Q13. User requirements are expressed as Scenarios in Extreme Programming.

- A. True
- B. False

Answer: A) True

Explanation:

User requirements are expressed as Scenarios in Extreme Programming.

Q14. It is depended to customer to customer to get involve in test development and validation in Extreme Programming.

- A. True
- B. False

Answer: A) True

Explanation:

It is depended to customer to customer to get involve in test development and validation in Extreme Programming.

Q15. The core functionality of scrum master is / are,

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- A. To improving the performance of the team
- B. To track and monitoring project development
- C. To obtain the project activity properly
- D. All of the mentioned above

Answer: D) All of the mentioned above

Explanation:

The core functionality of scrum master is to improving the performance of the team, to track and monitoring project development and To obtain the project activity properly.

Q.1

The most important feature of spiral model is

- (A) requirement analysis.
- (B) risk management.
- (C) quality management.
- (D) configuration management.

Ans: BQ.2

The worst type of coupling is

- (A) Data coupling.
- (B) control coupling.
- (C) stamp coupling.
- (D) content coupling.

Ans: DQ.3

IEEE 830-1993 is a IEEE recommended standard for

- (A) Software requirement specification.
- (B) Software design.
- (C) Testing.
- (D) Both

and

(B)Ans: AQ.4

One of the fault base testing techniques is

- (A)

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unit testing.

(B)

beta testing.

(C)

Stress testing.

(D)

mutation testing.

Ans: DQ.5

Changes made to an information system to add the desired but not necessarily therequired features is called

(A)

Preventative maintenance.

(B)

Adaptive maintenance.

(C)

Corrective maintenance.

(D)

Perfective maintenance.

Ans: DQ.6

All the modules of the system are integrated and tested as complete system in thecase of

(A)

Bottom up testing

(B)

Top-down testing

(C)

Sandwich testing

(D)

Big-Bang testing

Ans: D

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Q.7

If every requirement stated in the Software Requirement Specification (SRS) hasonly one interpretation, SRS is said to be

(A)

correct.

(B)

unambiguous.

(C)

consistent.

(D)

verifiable.

Ans: BQ.8

A fault simulation testing technique is

(A)
Mutation testing

(B)
Stress testing

(C)
Black box testing

(D)
White box testing

Ans: AQ.9

Modules X and Y operate on the same input and output data, then the cohesion is

(A)

Sequential

(B)
Communicational

(C)
Procedural

(D)
Logical

Ans: BQ.10

If the objects focus on the problem domain, then we are concerned with

(A)

Object Oriented Analysis.

(B)

Object Oriented Design

(C)

Object Oriented Analysis & Design

(D)

None of the above

Ans: AQ.11

SRS is also known as specification of

(A)

White box testing

(B)

Stress testing

(C)

Integrated testing

(D)

Black box testing

Ans: DQ.12

The model in which the requirements are implemented by category is

(A)

Evolutionary Development Model

(B)

Waterfall Model

(C)

Prototyping

(D)

Iterative Enhancement Model

Ans: AQ.13

SRD stands for

(A)

Software requirements definition

(B)

Structured requirements definition

(C)

Software requirements diagram

(D)

Structured requirements diagram

Ans: BQ.14

A COCOMO model is

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(A)

Common Cost Estimation Model.

(B)

Constructive Cost Estimation Model.

(C)

Complete Cost Estimation Model.

(D)

Comprehensive Cost Estimation Model.

Ans: BQ.15

Which of the following statements is true

(A)

Abstract data types are the same as classes

(B)

Abstract data types do not allow inheritance

(C)

Classes cannot inherit from the same base class

(D)

Object have state and behavior

Ans: BQ.16

The desired level of coupling is

(A)

No coupling

(B)

Control coupling

(C)

Common coupling

(D)

Data coupling

Ans: DQ.17

In the spiral model 'risk analysis' is performed

(A)

In the first loop

(B)

in the first and second loop

(C)

In every loop

(D)

before using spiral model

Ans: CQ.18

For a well understood data processing application it is best to use

(A)

The waterfall model

(B)

prototyping model

(C)

the evolutionary model

(D)

the spiral model

Ans: AQ.19

Coupling and cohesion can be represented using a

(A)

cause-effect graph

(B)

dependence matrix

(C)

Structure chart

(D)

SRS

Ans: BQ.20

The symbol represents

(A)

mandatory 1 cardinality

(B)

mandatory many cardinality

(C)

optional 0 or 1 cardinality

(D)

optional zero-many cardinality

Ans: DQ.21

Each time a defect gets detected and fixed, the reliability of a software product

(A)

increases.

(B)

decreases.

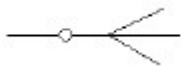
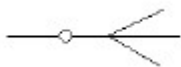
(C)

remains constant.

(D)

cannot say anything.

Ans: A



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1. Software is considered to be collection of _____.

- A. programming code
- B. associated libraries
- C. documentations
- D. All of the above

[View Answer](#)

Ans : D

Explanation: Software is considered to be collection of executable programming code, associated libraries and documentations.

2. The process of developing a software product using software engineering principles and methods is referred to as_____.

- A. Software Engineering
- B. software Evolution
- C. System Models
- D. Software Models

[View Answer](#)

Ans : B

Explanation: The process of developing a software product using software engineering principles and methods is referred to as software evolution.

3. Lehman has given laws for software evolution and he divided the software into _____ different categories.

- A. 6
- B. 2
- C. 3
- D. 5

[View Answer](#)

Ans : C

Explanation: Lehman has given laws for software evolution. He divided the software into three different categories: S-type, P-type and E-type.

4. Which of the following is not consider laws for E-Type software evolution?

- A. Continuing quality
- B. Continuing change
- C. Increasing complexity
- D. Self-regulation

[View Answer](#)

Ans : A

Explanation: Continuing quality is not consider laws for E-Type software evolution instead of that Reducing quality is the law.

5. Which of the following laws for E-Type says "E-type software system must continue to adapt to the real world changes, else it becomes progressively less useful".

- A. Continuing growth
- B. Continuing change
- C. Conservation of familiarity
- D. Self-regulation

[View Answer](#)

Ans : B

Explanation: Continuing change : An E-type software system must continue to adapt to the real world changes, else it becomes progressively less useful.

6. Which of the following is the Characteristics of good software?

- A. Transitional
- B. Operational
- C. Maintenance
- D. All of the above

[View Answer](#)

Ans : D

Explanation: A software product can be judged by what it offers and how well it can be used. This software must satisfy on the following grounds: Operational, Transitional, Maintenance.

7. Where there is a need of Software Engineering?

- A. For Large Software
- B. To reduce Cost
- C. Software Quality Management
- D. All of the above

[View Answer](#)

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Ans : D

Explanation: The need of software engineering arises because of higher rate of change in user requirements and environment on which the software is working.

8. The reason for software bugs and failures is due to _____.

- A. Software Developers
- B. Software companies
- C. Both A and B
- D. None of the above

View Answer

Ans : C

Explanation: Software companies are responsible for making policies and providing working atmosphere for the software development, so in turn these companies become a part of software development process. Bugs from developers side is no new thing. Thus option c answers the question.

9. Efficiency in a software product does not include _____

- A. licensing
- B. processing time
- C. responsiveness
- D. memory utilization

View Answer

Ans : A

Explanation: Licensing of a software product comes under corporate part of the software company.

10. What are attributes of good software ?

- A. Software functionality
- B. Software development
- C. Software maintainability
- D. Both A and C

View Answer

Ans : D

Explanation: Good software should deliver the required functionality, maintainability. Software development is not an attribute but a fundamental.

01. SDLC stands for _____

- A. Software development life cycle
- B. Software design life cycle
- C. System design life cycle
- D. System development life cycle

View Answer

Answer : A

Explanation: SDLC stands for Software development life cycle. Software Development Life Cycle (SDLC) is a process used by the software industry to analyze user requirements and then design, develop, test and maintain high quality softwares. The SDLC aims to produce a high-quality software that meets or exceeds customer expectations, reaches completion within times and cost estimates. So, option A is correct.

02. The spiral model was originally proposed by

- A. IBM
- B. Barry Boehm
- C. Royce
- D. Pressman

View Answer

Answer : B

Explanation: The spiral model was originally proposed by Barry Boehm. So, option B is correct.

03. Agile Software Development is based on

- A. Incremental Development
- B. Iterative Development
- C. Both Incremental and Iterative Development
- D. Linear Development

View Answer

Answer : C

Explanation: In agile Software Development, the software is developed in increments with the customer specifying the requirements to be included in each increment and the highest priority is to satisfy the customer through early and continuous delivery of valuable software. They are iterative because they work on one iteration followed by improvements in next iteration and so on. So, option C is correct.

04. Which of the following is a functional requirement

- A. Portability
- B. Robustness
- C. Maintainability
- D. None of the above

View Answer

Answer : D

Explanation: All of the above mentioned are non-functional requirement which represents the quality of the software. So, option D is correct.

05. What is the major advantage of using the Incremental Model?

- A. Easier to test and debug and it is used when there is a need to get a product to the market early
- B. Customer can respond to each increment
- C. It is used when there is a need to get a product to the market early
- D. Easier to test and debug

View Answer

Answer : D

Explanation: Major advantage of using the Incremental Model is that it is generally easier to test and debug than other methods of software development because relatively smaller changes are made during each iteration and is popular particularly when we have to quickly deliver a limited functionality system. So, option A is correct.

06. Selection of a model is based on

- A. Requirements
- B. Development team
- C. Users
- D. All of the mentioned

View Answer

Answer : D

Explanation: Selection of a model is based on all of the above mentioned: Requirements, Development team and Users.

07. Purpose of process is to deliver software

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- A. In time
- B. With acceptable quality
- C. That is cost efficient
- D. Both in time & with acceptable quality

View Answer

Answer : D

Explanation: Purpose of process is to deliver software Both in time & with acceptable quality.

08. Which one of the following is not a software process quality?

- A. Portability
- B. Productivity
- C. Timeliness
- D. Visibility

View Answer

Answer : A

Explanation: Portability is not a software process quality. So, option A is correct.

09. What are the attributes of good software?

- A. Software maintainability
- B. Software maintainability & functionality
- C. Software functionality
- D. Software development

View Answer

Answer : B

Explanation: Good software should deliver the required functionality and maintainability however, Software development is not an attribute but a fundamental. So, option B is correct.

10. Efficiency in a software product does not include _____

- A. Processing time
- B. Responsiveness
- C. Licensing
- D. Memory utilization

View Answer

Answer : C

Explanation: Efficiency in a software product does not include licensing.

11. What is the first step in the software development lifecycle?

- A. Preliminary Investigation and Analysis
- B. System Testing
- C. System Design
- D. Coding

View Answer

Answer : A

Explanation: The first phase of the software development life cycle is the requirement analysis, in which the software engineer analyzes the software requirement according to the user so that the user will get the final product according to their need.

12. In the maintenance phase the product must be tested against previous test cases. This is known as _____ testing

- A. Unit

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- B. Regression
- C. Integration
- D. Module

View Answer

Answer : B

Explanation: In the maintenance phase the product must be tested against previous test cases and this is known as regression testing.

13. The reason for software bugs and failures is due to _____

- A. Software Developers
- B. Software companies
- C. Both A and B
- D. None of the above

View Answer

Answer : C

Explanation: Software companies are responsible for making policies and providing working atmosphere for the software development, so in turn these companies become a part of software development process. Bugs from developers side is no new thing. So, option C is correct.

14. What is the simplest model of software development paradigm?

- A. V-model
- B. Big bang model
- C. Spiral model
- D. Waterfall model

View Answer

Answer : D

Explanation: Waterfall model is the simplest model of software development paradigm. Thus option D is correct.

15. Which of these are the 5 generic software engineering framework activities?

- A. Communication, planning, modeling, construction, deployment
- B. Communication, risk management, measurement, production, reviewing
- C. Analysis, designing, programming, debugging, maintenance
- D. Analysis, planning, designing, programming, testing

View Answer

Answer : A

Explanation: The 5 generic software engineering framework activities are: communication, planning, modeling, construction and deployment. Option A is correct.

16. What does the study of an existing system refer to?

- A. Communication, planning, modeling, construction, deployment
- B. Communication, risk management, measurement, production, reviewing
- C. Analysis, designing, programming, debugging, maintenance
- D. Analysis, planning, designing, programming, testing

View Answer

Answer : A

Explanation: The 5 generic software engineering framework activities are: communication, planning, modeling, construction and deployment. Option A is correct.

17. Software Engineering is a:

- A. Complicated way to develop software
- B. Systematic way to develop software

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- C. Cost-effective way to develop software
- D. Both B and C

View Answer

Answer : D

Explanation: Software Engineering is a systematic and cost-effective way to develop software. So both statement B and C are correct.

18. Which of the following is/are considered stakeholder in the software process?

- A. Customers
- B. Project managers
- C. End-users
- D. All of the above

View Answer

Answer : D

Explanation: All of the above (Customers, Project managers, End-users) are considered stakeholder in the software process.

19. Which of the items listed below is not one of the software engineering layers?

- A. Process
- B. Manufacturing
- C. Methods
- D. Tools

View Answer

Answer : B

Explanation: None

20. What is the most common measure for correctness?

- A. Errors per KLOC
- B. Defects per KLOC
- C. \$ per KLOC
- D. Pages of documentation per KLOC

View Answer

Answer : B

Explanation: The most common measure for correctness is Defects per KLOC. So, option B is correct

21. Which of the following is involved in the system planning and designing phase of the Software Development Life Cycle (SDLC)?

- A. Specification freeze
- B. Sizing
- C. Parallel run
- D. All of the above

View Answer

Answer : D

Explanation: Sizing, Parallel run, Specification freeze all of the above are involved in the system planning and designing phase of the Software Development Life Cycle (SDLC)

22. Which of the following projects would be a good one for adopting the prototyping paradigm for software development?

- A. Algebra Tutor
- B. Accounting System
- C. Spreadsheet

D. Automobile Cruise Control

View Answer

Answer : A

Explanation: Algebra Tutor would be a good one for adopting the prototyping paradigm for software development. So, option A is correct.

23. One of the fault-based testing techniques is

- A. Unit Testing
- B. Beta Testing
- C. Mutation Testing
- D. Stress Testing

View Answer

Answer : C

Explanation: Mutation Testing is one of the fault-based testing techniques.

24. What is the major drawback of the Spiral Model?

- A. Higher amount of risk analysis
- B. Doesn't work well for smaller projects
- C. Additional functionalities are added later on
- D. Strong approval and documentation control

View Answer

Answer : B

Explanation: The major drawback of the Spiral Model is that it doesn't work well for smaller projects.

25. Which of the following is not consider laws for E-Type software evolution?

- A. Continuing change
- B. Continuing quality
- C. Self-regulation
- D. Increasing complexity

View Answer

Answer : B

Explanation: Continuing quality is not consider laws for E-Type software evolution instead of that Reducing quality is the law.

26. Process models are described as agile because they

- A. Eliminate the need for cumbersome documentation
- B. Emphasize maneuverability and adaptability
- C. Do not waste development time on planning activities
- D. Make extensive use of prototype creation

View Answer

Answer : B

Explanation: Emphasize maneuverability and adaptability is the correct answer.

27. When elements of module are grouped together that are executed sequentially in order to perform a task, is called _____

- A. Logical cohesion
- B. Procedural cohesion
- C. Co-incidental cohesion
- D. Emporal cohesion

View Answer

Answer : B

Explanation: When elements of module are grouped together that are executed sequentially in order to perform a task, is called Procedural Cohesion.

28. Which model is also known as Verification and validation model?

- A. V-model
- B. Waterfall model
- C. Big Bang model
- D. Spiral model

View Answer

Answer : A

Explanation: None

29. UML diagram cannot be used to capture _____ views

- A. Users view
- B. Structural view
- C. Layer view
- D. Environmental view

View Answer

Answer : C

Explanation: Layer view is the correct answer. Other options are captured using UML diagram. So, option C is correct.

30. Which model is not suitable for large software projects but good one for learning and experimenting?

- A. Big Bang model
- B. Iterative model
- C. Spiral model
- D. Waterfall model

View Answer

Answer : A

Explanation: Big Bang model is the correct answer.

1. Which of these does not affect different types of software as a whole?

- a. Security
- b. Business and social change
- c. Flexibility
- d. Heterogeneity

Answer: (c) Flexibility

2. The 4GT Model acts as a package of _____.

- a. Software Tools
- b. Software Programs

- c. CASE tools
- d. None of the above

Answer: (a) Software tools

3. _____ originally proposed the spiral model.

- a. Pressman
- b. Royce
- c. IBM
- d. Barry Boehm

Answer: (d) Barry Boehm

4. Out of the following, which one is not a workbench tool for Requirement Management?

- a. RDD 100
- b. Rational Suite
- c. DOORS
- d. RTM

Answer: (b) Rational Suite

5. In system modelling, which model depicts a system's static nature?

- a. Data Model
- b. Structural Model
- c. Context Model
- d. Behavioural Model

Answer: (b) Structural Model

6. Source code translation and Program modularisation are the activities of _____.

- a. Reverse engineering
- b. Forward Engineering
- c. Reengineering and Reverse Engineering
- d. Reengineering

Answer: (d) Reengineering

7. Which of these reliability requirements is concerned with maintaining various system copies?

- a. Ambiguous Requirements
- b. Redundancy Requirements
- c. Recovery Requirements
- d. Checking Requirements

Answer: (c) Recovery Requirements

8. The worst form of module cohesion is_____.

- a. Coincidental Cohesion
- b. Functional Cohesion
- c. Temporal Cohesion
- d. Logical Cohesion

Answer: (a) Coincidental Cohesion

9. How does the implementation of generalisation occur in Object-Oriented programming languages?

- a. Encapsulation
- b. Abstract Classes
- c. Inheritance
- d. Polymorphism

Answer: (c) Inheritance

10. Out of these early OOD methods, which one incorporates both a “macro development process” and a “micro development process.”?

- a. Wirfs-Brock method
- b. Booch method
- c. Coad and Yourdon method
- d. Rumbaugh method

Answer: (b) Booch Method

11. Out of these web engineering metrics, which one measures the relatedness extent between multiple web pages?

- a. Static Content Objects Numbers
- b. Dynamic Content Objects Numbers
- c. Internal Page Links Number
- d. Web Page Similarity

Answer: (d) Web Page Similarity

12. Which one isn't a stage of the COCOMO-II?

- a. Model of Application Composition estimation
- b. Model of Early design estimation
- c. Model of Post architecture estimation
- d. Model of Comprehensive cost estimation

Answer: (b) Model of Early design estimation

13. Which one isn't an approach to estimate the cost of the software?

- a. Heuristic
- b. Empirical

- c. Critical
- d. Analytical

Answer: (c) Critical

14. The aviation industry follows which of these standards?

- a. RTCADO-178B
- b. CTRADO-172B
- c. CTCADO-178B
- d. RTRADO-178B

Answer: (a) RTCADO-178B

15. We generally use the _____ for Software Maintenance.

- a. Integration Testing
- b. Unit Testing
- c. System Testing
- d. Regression Testing

Answer: (d) Regression Testing

16. The language-processing system performs the translation of an XML data description to a_____.

- a. alternative XML representation
- b. machine code
- c. software module
- d. alternative XML representation and machine code

Answer: (d) alternative XML representation and machine code

17. What establishes the profile of end-users of the system?

- a. system image
- b. mental image
- c. user's model
- d. design model

Answer: (c) user's model

18. Which is a stimuli category?

- a. Management stimuli
- b. Hardware stimuli
- c. Software stimuli
- d. Periodic stimuli

Answer: (d) Periodic stimuli

19. An object does NOT provide this service:

- a. Deactivating & Activating Objects
- b. Files implementing the identified entities in the ERD
- c. Registering object implementation
- d. Security features

Answer: (b) Files implementing the identified entities in the ERD

20. Out of these, which one isn't a business process's example?

- a. purchasing services
- b. testing software
- c. designing a new product
- d. hiring an employee

Answer: (b) testing software

21. Which one of these is a data problem?

- a. record organisation problem
- b. hardware problems
- c. loss of comments
- d. heavy computational demands

Answer: (a) record organisation problems

22. The FIT (Failure In Time) is another way in which one can REPORT _____.

- c. MTBF
- d. MTSF
- a. MTTF
- b. MTTR

Answer: (a) MTBF

23. What do we mean by Six Sigma?

- a. A widely used strategy to implement statistical quality assurance
- b. Six standard deviations
- c. A widely used strategy that maintains statistical quality assurance and six standard deviations
- d. An FTR (Formal Technical Review) guideline for inspection or a quality walkthrough

Answer: (c) A widely used strategy that maintains statistical quality assurance and six standard deviations

24. Name an evaluation technique to assess the quality of test cases.

- a. Validation
- b. Mutation analysis

c. Performance analysis

d. Verification

Answer: (b) Mutation analysis

25. Code is checked in which type of testing?

a. White box testing

b. Black box testing

c. Green box testing

d. Red box testing

Answer: (a) White box testing